

QAOA with ideal noise returns wrong result

<https://github.com/qiskit-community/qiskit-aqua/issues/1509>

ROW 59

QAOA with ideal noise returns wrong result #1509



Closed



iuliazidaru opened on Jan 18, 2021 · edited by iuliazidaru

Edits ...

Information

- Qiskit Aqua version: qiskit-aqua: '0.8.1'
- Python version: 3.8.2
- Operating system: System: Darwin

What is the current behavior?

QAOA applied to the linear equation: minimize obj($x_1 + x_2 + x_3 + \dots + x_{20}$) returns wrong result: 2

Steps to reproduce the problem

```
##%%

from qiskit.optimization import QuadraticProgram
def create_qubo_linear(num_variables):
    qubo = QuadraticProgram()
    for index in range(num_variables):
        qubo.binary_var('x' + str(index + 1))
    qubo.minimize(linear=[1] * num_variables,
                  quadratic={})
    return qubo

##%%

from qiskit.aqua import QuantumInstance
from qiskit.aqua import aqua_globals
from qiskit.optimization.algorithms import MinimumEigenOptimizer
from qiskit.aqua.algorithms import QAOA

SEED = 10598

def solve_qubo(qubo, device, optimiser, parameter_p):
    aqua_globals.random_seed = SEED
    quantum_instance = QuantumInstance(device, seed_simulator=SEED, seed_transpiler=SEED, shots=1024)
    qaoa_mes = QAOA(quantum_instance=quantum_instance, include_custom=True, optimizer=optimiser,
                    p=parameter_p)
    qaoa = MinimumEigenOptimizer(qaoa_mes)
    qaoa_result = qaoa.solve(qubo)
    return qaoa_result

##%%

from qiskit.aqua import aqua_globals
from qiskit.aqua.components.optimizers import SLSQP
from qiskit.providers.aer import QasmSimulator
from qiskit.providers.aer.noise import NoiseModel

def test_given_qubo_linear_then_qaoa_correct_result():
    # Arrange
    nbr_variables = 20
    qubo = create_qubo_linear(nbr_variables)
    noise_model = NoiseModel()
    ##device = QasmSimulator(method='matrix_product_state')
    ##device = QasmSimulator(method='density_matrix') # insufficient memory
    device = QasmSimulator(method='automatic', noise_model=noise_model)
    #device = Aer.get_backend('qasm_simulator')
    # Act
    result = solve_qubo(qubo, device, SLSQP(), 5)
    # Assert
    print(noise_model.is_ideal())
    print(result.fval)
    print(result.x)
    assert noise_model.is_ideal() == True
    assert result.fval == 0.0
    assert all([a == b for a, b in zip(result.x, [0] * nbr_variables)])

##%%

test_given_qubo_linear_then_qaoa_correct_result()

##%%
```



Assignees

No one assigned

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No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

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Participants



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```
import sys
print(sys.version)

#95%

import platform
uname = platform.uname()
print(f"System: {uname.system}")
print(f"Node Name: {uname.node}")
print(f"Release: {uname.release}")
print(f"Version: {uname.version}")
print(f"Machine: {uname.machine}")
print(f"Processor: {uname.processor}")

#95%
```

What is the expected behavior?

minimum value must be 0, and the solution 00000000000000000000

Suggested solutions

I replaced the problem in QAOA example <https://qiskit.org/textbook/ch-applications/qaoa.html> and the result looks good.

```
n = 20
V = np.arange(0,n,1)
E = []
```

[QAOA-algorithm.ipynb.zip](#)
[QAOA-bug.ipynb.zip](#)

iuliazidaru on Jan 18, 2021

Author ...

This is related to : [#1434](#)

 Juliazidaru changed the title **QAOA with ideal noise return wrong result** **QAOA with ideal noise returns wrong result** on Jan 25, 2021

manoelmarques on Jan 26, 2021

Collaborator ...

In the above notebook just changing from SLSQP() to COBYLA(), I got the correct results.

iuliazidaru on Jan 27, 2021

Author ...

Thank you for the hint [@manoelmarques](#) !

I have 3 tests for the same scenario in 3 projects. The test I created in qiskit-aqua worked with COBYLA. Using

```
quantum_instance = QuantumInstance(QasmSimulator(method='automatic', noise_model=NoiseModel()),
                                   seed_simulator=aqua_globals.random_seed,
                                   seed_transpiler=aqua_globals.random_seed,
                                   shots=1024)
```

I get a solution sample containing multiple values and value 0.

[illegible]

